Laboratory assignment

Component (2) – Data analysis

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1 Analysis of the features used in learning

1.1 Correlation

Table 1: Pairs of features by strength of correlation.

	Pearson		Spearman	
Condition	pairs	%	pairs	%
$\lvert r\rvert \geq 0.5$	64 879	41.3	85 093	54.2
$\lvert r\rvert \geq 0.8$	22 815	14.5	61 248	39.0
$\lvert r\rvert \geq 0.9$	8 093	5.2	27 130	17.3
$\lvert r\rvert \geq 0.95$	2 219	1.4	4 249	2.7
$\lvert r\rvert \geq 0.99$	207	0.1	492	0.3
$\lvert r\rvert < 0.2$	51 275	32.6	40 444	25.7

Table 2: Pairs of features by band of correlation.

	Pearson		Spearman	
Band	pairs	%	pairs	%
$0 \leq \lvert r\rvert < 0.2$	51 275	32.6	40 444	25.7
$0.2 \leq \lvert r\rvert < 0.5$	40 926	26.1	31 543	20.1
$0.5 \leq \lvert r\rvert < 0.8$	42 064	26.8	23 845	15.2
$0.8 \leq \lvert r\rvert < 0.9$	14 722	9.4	34 118	21.7
$0.9 \leq \lvert r\rvert \leq 1$	8 093	5.2	27 130	17.3

Table 3: Notable pairs of distinct features.

Feature 1	Feature 2	Pearson	Spearman
`tBodyAccJerk-energy()-X`	`fBodyAccJerk-energy()-X`	1.000	1.000
`tGravityAcc-arCoeff()-Z,1`	`tGravityAcc-arCoeff()-Z,2`	−0.995	−0.995
`tGravityAcc-mean()-Z`	`angle(Z,gravityMean)`	−0.995	−1.000
`fBodyGyro-bandsEnergy()-1,8_2`	`fBodyGyro-bandsEnergy()-17,24_2`	0.273	0.881

- Median $\lvert r\rvert$: Pearson 0.38, Spearman 0.58.
- 19 groups of identical features, 21 redundant copies: Table 4.

Table 4: Groups of identical features.

Group	Features
1	tBodyAccMag-mean(), tBodyAccMag-sma(), tGravityAccMag-mean(), tGravityAccMag-sma()
2	tBodyAccMag-std(), tGravityAccMag-std()
3	tBodyAccMag-mad(), tGravityAccMag-mad()
4	tBodyAccMag-max(), tGravityAccMag-max()
5	tBodyAccMag-min(), tGravityAccMag-min()
6	tBodyAccMag-energy(), tGravityAccMag-energy()
7	tBodyAccMag-iqr(), tGravityAccMag-iqr()
8	tBodyAccMag-entropy(), tGravityAccMag-entropy()
9	tBodyAccMag-arCoeff()1, tGravityAccMag-arCoeff()1
10	tBodyAccMag-arCoeff()2, tGravityAccMag-arCoeff()2
11	tBodyAccMag-arCoeff()3, tGravityAccMag-arCoeff()3
12	tBodyAccMag-arCoeff()4, tGravityAccMag-arCoeff()4
13	tBodyAccJerkMag-mean(), tBodyAccJerkMag-sma()
14	tBodyGyroMag-mean(), tBodyGyroMag-sma()
15	tBodyGyroJerkMag-mean(), tBodyGyroJerkMag-sma()
16	fBodyAccMag-mean(), fBodyAccMag-sma()
17	fBodyBodyAccJerkMag-mean(), fBodyBodyAccJerkMag-sma()
18	fBodyBodyGyroMag-mean(), fBodyBodyGyroMag-sma()
19	fBodyBodyGyroJerkMag-mean(), fBodyBodyGyroJerkMag-sma()

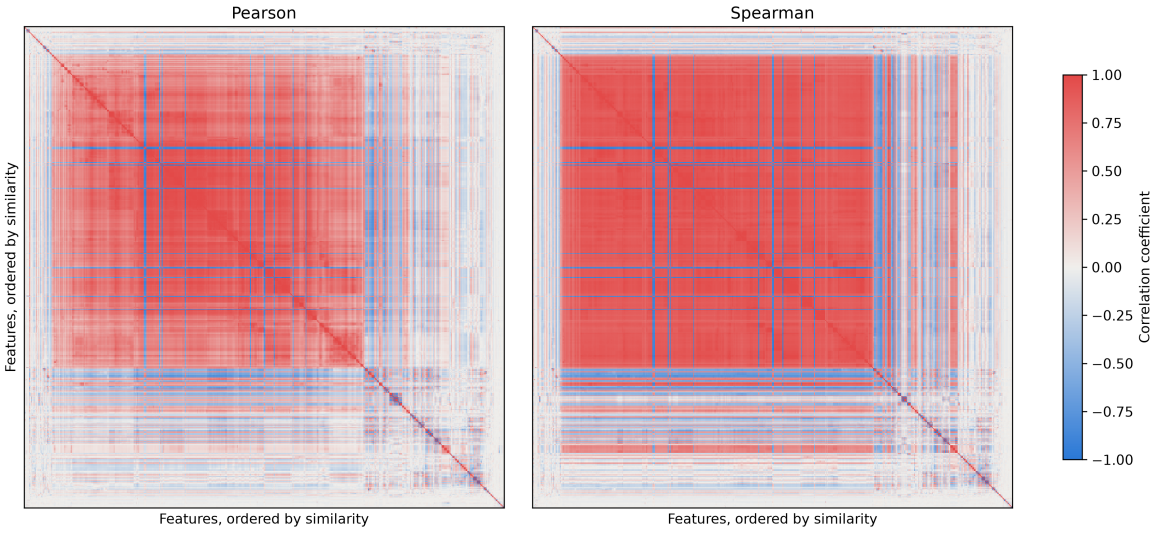


Figure 1: Correlation matrices; features ordered by hierarchical clustering on the Pearson matrix, same order in both panels.

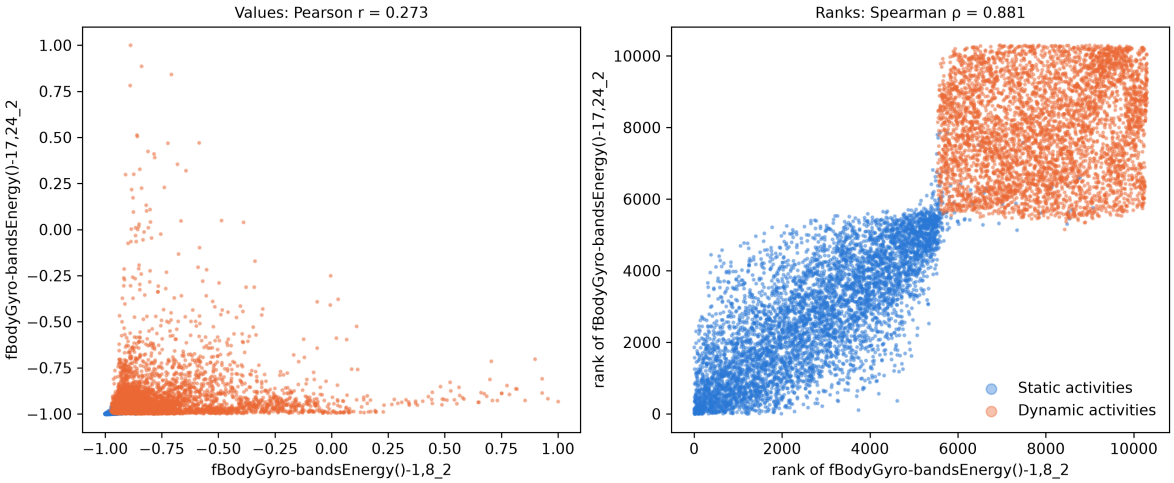


Figure 2: `fBodyGyro-bandsEnergy()-1,8_2` vs. `fBodyGyro-bandsEnergy()-17,24_2`: values and ranks, static and dynamic activities.

- $|\rho| \geq 0.8$ and $|r| < 0.5$: 3 983 pairs; high ρ mainly from the static vs. dynamic split; within dynamic activities, median $|\rho| = 0.23$; e.g. last row of Table 3, Figure 2.
- `std`, `mad`, `iqr`, `max` of the same signal: strongly correlated. Example, Pearson r:
  - `tBodyAcc-std()-X` – `tBodyAcc-mad()-X`: 0.999
  - `tBodyAcc-std()-X` – `tBodyAcc-iqr()-X`: 0.981
  - `tBodyAcc-std()-X` – `tBodyAcc-max()-X`: 0.981

1.2 Independence

Pearson's χ^2 test on binned features, Bonferroni-corrected $\alpha = 0.05$.

Table 5: Results of the χ^2 tests.

	Feature vs. activity	Feature vs. feature
Features	561	540 (distinct)
Bins	10	5
Tests	561	145 530
Significance level	8.9×10^{-5}	3.4×10^{-7}
Independence rejected	561 (100%)	144 190 (99.1%)
Cramér's V: min / median / max	0.058 / 0.447 / 0.570	0.012 / 0.339 / 1.000
$V \geq 0.5$	46 (8.2%)	47 393 (32.6%)
$V < 0.1$	6 (1.1%)	12 822 (8.8%)
Features with fewer bins (repeated values)	20	17

Table 6: Features most and least dependent on the activity.

Most dependent	V	Least dependent	V
`fBodyAcc-bandsEnergy()-1,8_1`	0.570	`tBodyGyroJerk-arCoeff()-X,4`	0.058
`fBodyAcc-bandsEnergy()-1,16_1`	0.561	`tBodyAccJerkMag-arCoeff()3`	0.080
`fBodyAccMag-mad()`	0.558	`tBodyGyroJerk-arCoeff()-Y,4`	0.081
`fBodyAcc-std()-X`	0.558	`tBodyGyroJerkMag-arCoeff()3`	0.083
`fBodyAccMag-std()`	0.557	`tBodyAcc-arCoeff()-Y,4`	0.085

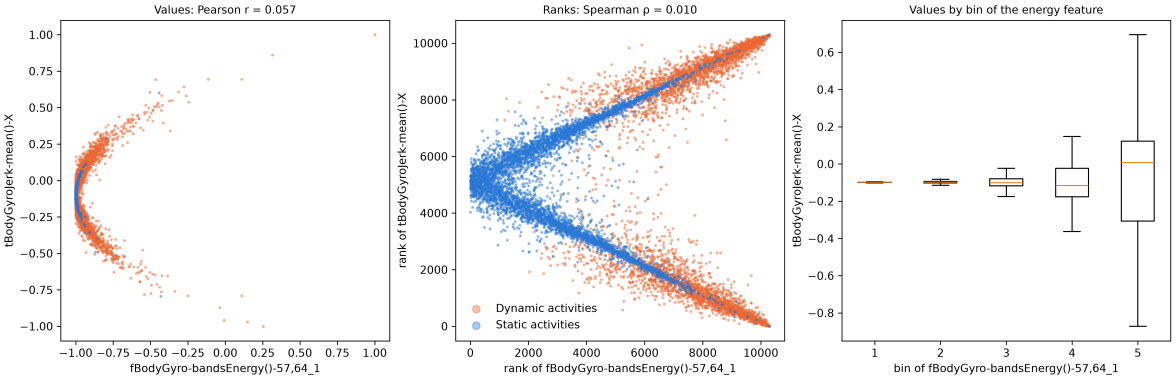


Figure 3: Uncorrelated but dependent features: values and ranks for static and dynamic activities, and box plots by bin of the energy feature.

- Uncorrelated does not mean independent: 19 605 pairs have $|r| < 0.1$ and $|\rho| < 0.1$; the χ^2 test finds 18 271 of them dependent, i.e. 93.2%.
- Example, Figure 3: `tBodyGyroJerk-mean()-X` and `fBodyGyro-bandsEnergy()-57,64_1`: $r = 0.057$, $\rho = 0.010$, $V = 0.580$.
- Windows overlap by 50%: instances not independent, p-values optimistic.

1.3 Feature importance

Table 7: Top 10 features by each measure.

#	Feature	F	IG (bits)
Ranked by F			
1	`fBodyAccJerk-entropy()-X`	50776	0.991
2	`tGravityAcc-mean()-X`	46277	0.840
3	`tGravityAcc-min()-X`	44358	0.910
4	`tGravityAcc-max()-X`	41203	0.777
5	`tGravityAcc-energy()-X`	40416	0.840
6	`fBodyAccJerk-entropy()-Y`	37518	0.972
7	`fBodyBodyAccJerkMag-entropy()`	32616	0.997
8	`tBodyAccJerkMag-entropy()`	32508	1.022
9	`fBodyAcc-entropy()-X`	32201	1.020
10	`tBodyAccJerk-entropy()-X`	31258	0.964
Ranked by IG			
1	`fBodyAcc-bandsEnergy()-1,8_1`	7083	1.170
2	`fBodyAcc-bandsEnergy()-1,16_1`	7598	1.152
3	`fBodyAcc-std()-X`	22770	1.144
4	`tBodyAcc-max()-X`	27862	1.142
5	`fBodyAcc-bandsEnergy()-1,24_1`	7841	1.139
6	`tBodyAcc-std()-X`	24559	1.135
7	`fBodyAcc-energy()-X`	7875	1.135
8	`tBodyAcc-energy()-X`	7756	1.134
9	`fBodyAcc-max()-X`	13854	1.125
10	`tBodyAcc-mad()-X`	21278	1.117

- $\max IG = 1.170$ bits, of $H = 2.576$ bits.
- Rank correlation F vs. IG: 0.789; common features in the two top 20: 4.
- Median number of features with $|r| \geq 0.9$ within the same top 20: F 13.5, IG 17.

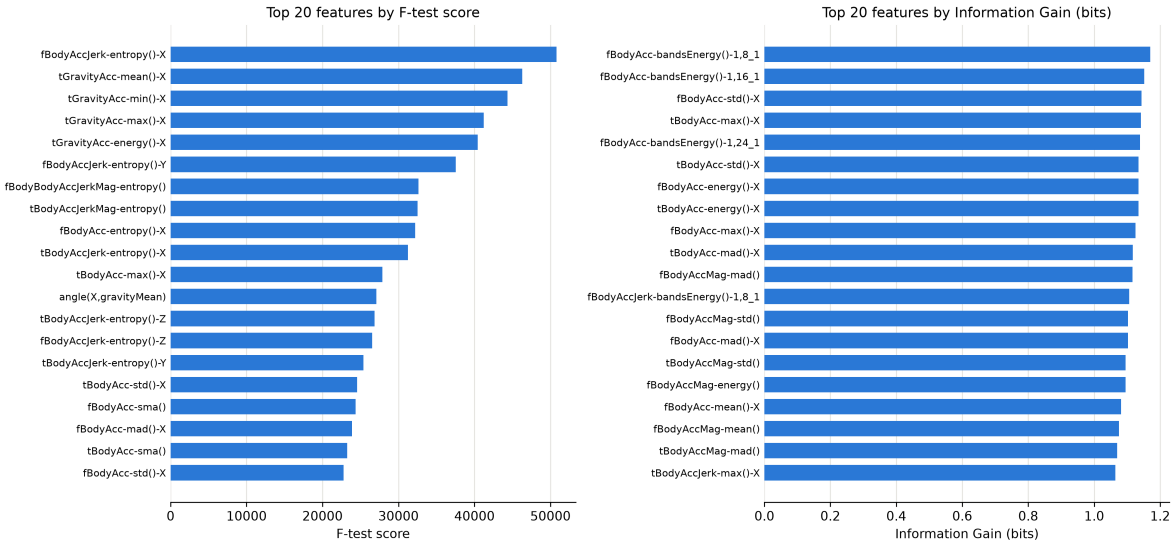


Figure 4: Top 20 features by F-test score and by Information Gain.

Table 8: Median importance by feature family.

Statistic	n	F	IG	Group	n	F	IG
`sma`	8	15857	0.991	*Sensor*			
`energy`	32	3079	0.978	body acceleration	286	2517	0.938
`mad`	32	9391	0.971	gyroscope	207	1475	0.905
`std`	32	8626	0.964	angle	7	44	0.462
`iqr`	32	7609	0.959	gravity	40	519	0.419
`max`	32	6462	0.947	*Domain*			
`entropy`	32	13555	0.945	frequency	285	1863	0.934
`bandsEnergy`	126	1634	0.939	time	248	2177	0.675
`mean`	32	6795	0.916	angle	7	44	0.462
`min`	32	3706	0.878				
`angle`	7	44	0.462				
`maxInds`	13	272	0.340				
`meanFreq`	13	473	0.234				
`skewness`	13	268	0.184				
`arCoeff`	76	388	0.157				
`kurtosis`	13	247	0.153				
`correlation`	15	186	0.141				

2 Data statistics

Table 9: Data set overview: UCI HAR [AGO+13], training and test files.

Instances	10 299	Volunteers	30
Training / test instances	7 352 / 2 947	Activities	6
Features	561	Missing values	0
Distinct features	540	Repeated rows	0

Table 10: Instances per activity.

	Walking	Upstairs	Downstairs	Sitting	Standing	Laying
Instances	1 722	1 544	1 406	1 777	1 906	1 944
Share (%)	16.7	15.0	13.7	17.3	18.5	18.9

- Class balance: largest / smallest class = 1944/1406 = 1.38.
- Instances per volunteer: min = 281, max = 409, median = 342.5.
- Instances per volunteer and activity: min = 36, max = 95, median = 54.5; all 30 volunteers performed all 6 activities.

3 Data distribution

Studied for the 6 features with the highest Information Gain, Table 7, skipping features with $|r| \geq 0.9$ to an already selected one.

Table 11: Selected features: importance, shape, outliers and training vs. test difference.

Feature	IG	F	Skewness	Kurtosis	Outliers (%)	KS
`fBodyAcc-bandsEnergy()-1,8_1`	1.170	7083	1.91	4.12	5.86	0.055
`tBodyAcc-max()-X`	1.142	27862	0.60	−1.16	0.00	0.045
`tBodyAccJerk-energy()-X`	1.036	5001	1.80	4.31	3.75	0.065
`tBodyGyroJerk-mad()-X`	1.009	9494	0.86	−0.09	0.24	0.065
`fBodyAccJerk-bandsEnergy()-17,32_3`	1.007	1579	4.19	33.15	8.41	0.060
`fBodyAccJerk-bandsEnergy()-17,24_2`	0.999	2820	2.05	4.94	6.44	0.086

- Static activities: values ≤ -0.8 for 97.5–100% of instances, for all 6 features.
- Dynamic activities: spread over $[-1, 1]$; highest values for walking downstairs, e.g. `tBodyAcc-max()-X`: median = 0.459.
- All 6 features right-skewed: skewness $\in [0.60, 4.19]$; outliers $\in [0, 8.4]$%, mainly in the tails.
- Training vs. test: Kolmogorov–Smirnov statistic ≤ 0.086.

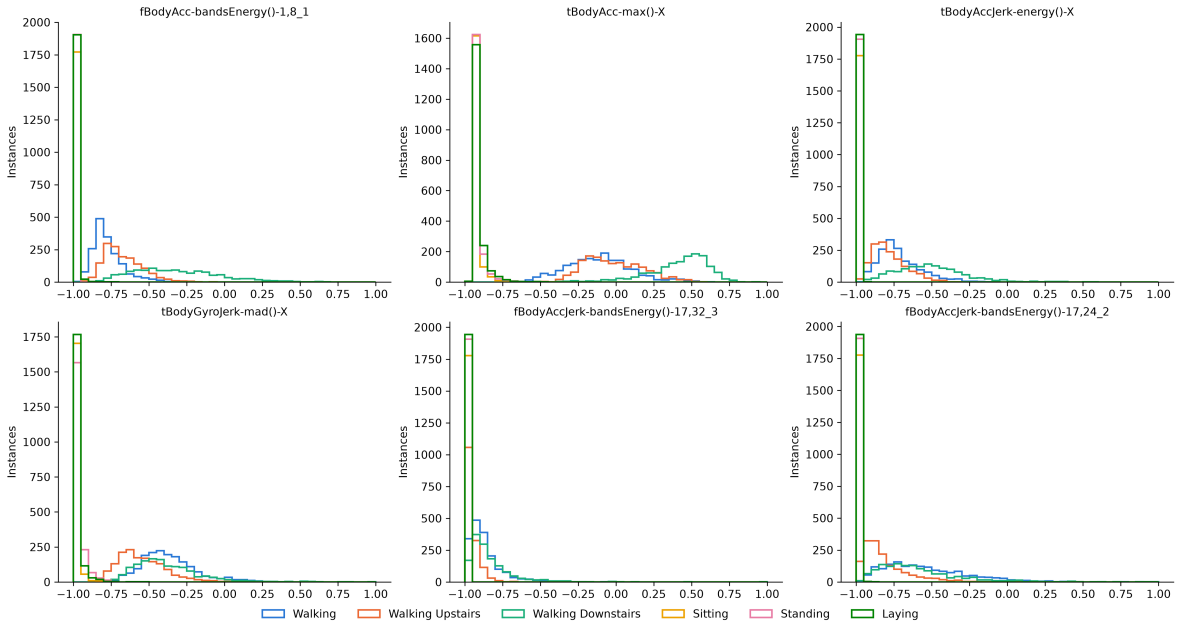


Figure 5: Histograms of the selected features, one line per activity.

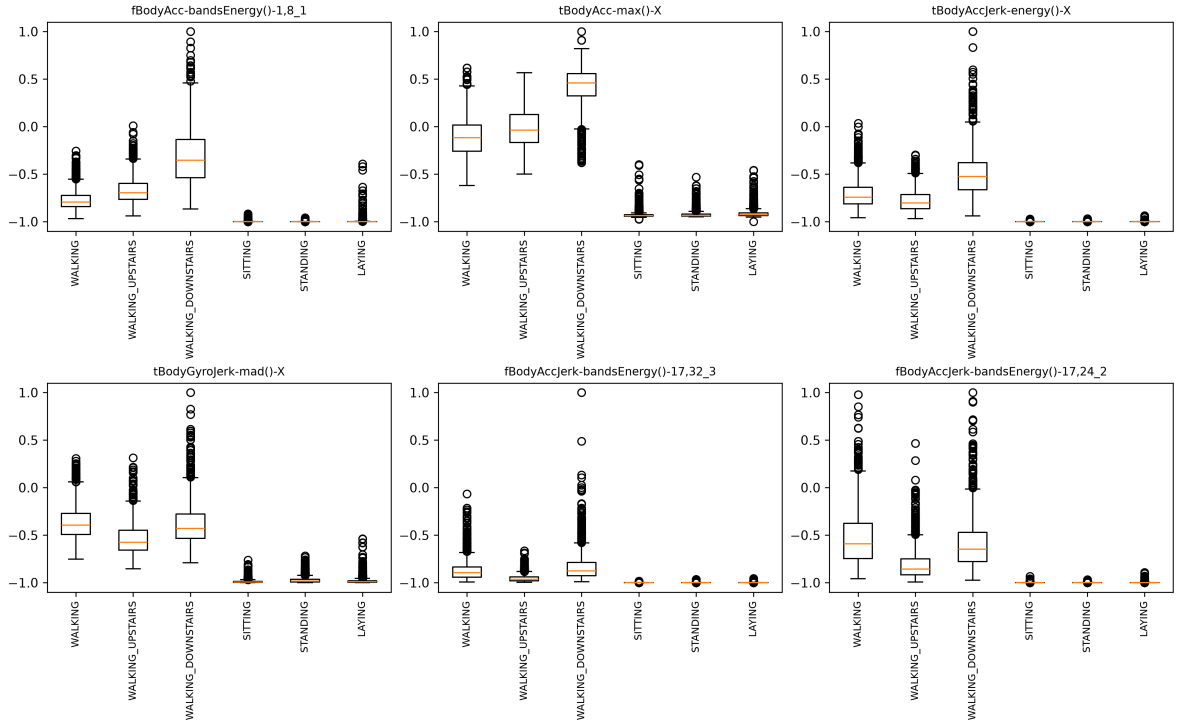


Figure 6: Box plots of the selected features by activity.

4 Data visualization and interpretation

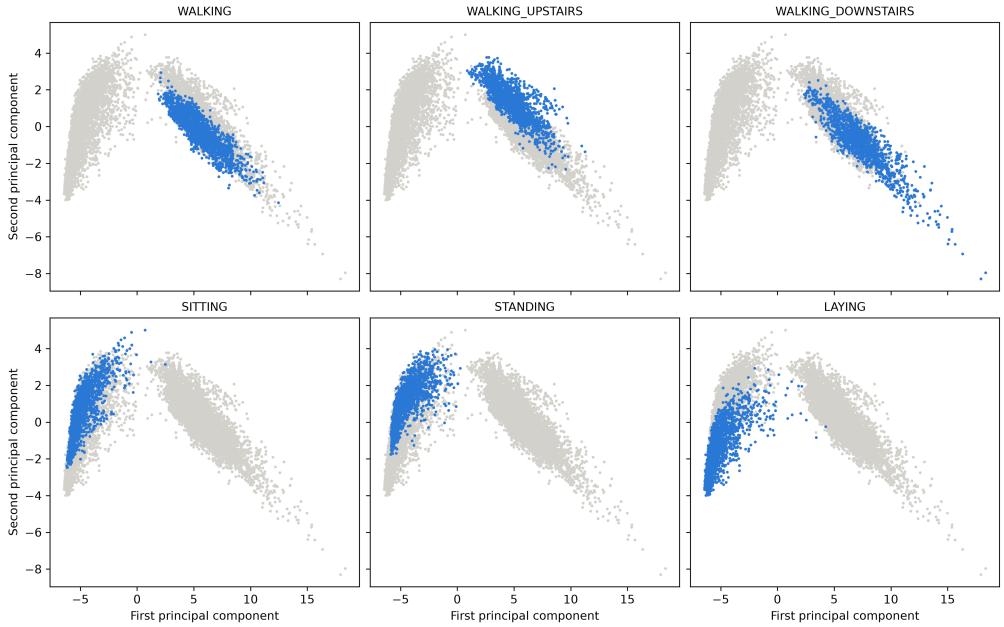


Figure 7: PCA projection of the 540 distinct features; explained variance: PC1 60.8%, PC2 4.8%. Blue: the activity of the panel; grey: all instances.

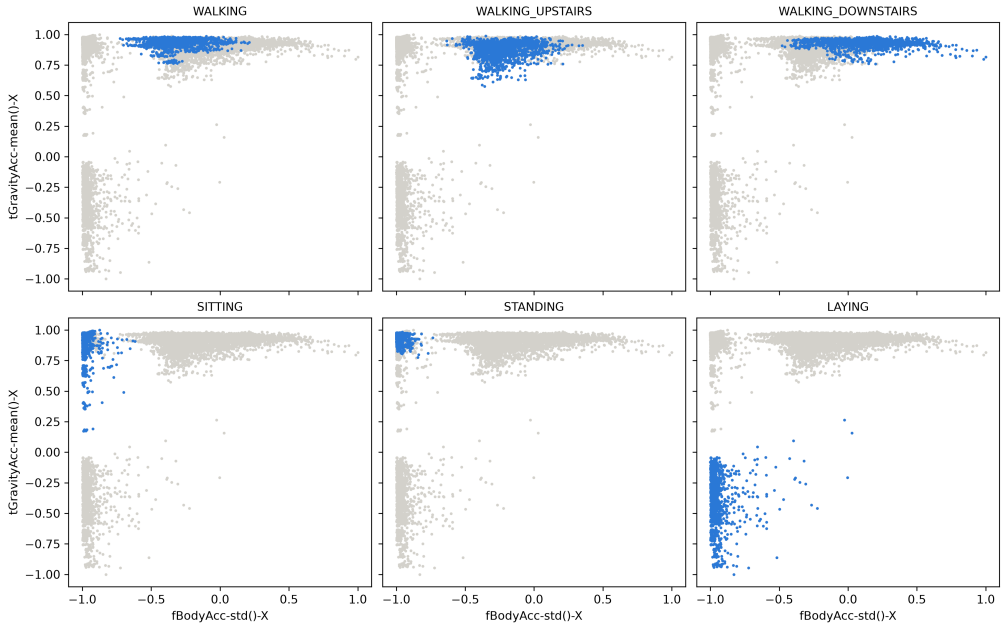


Figure 8: `fBodyAcc-std()-X` vs. `tGravityAcc-mean()-X`.

Interpretation:

- PCA, Figure 7: PC1 = 60.8%, PC2 = 4.8% of the variance.
  - PC1 separates the static from the dynamic activities into two groups.
  - Within each group, the activities overlap.
- Two features, Figure 8:

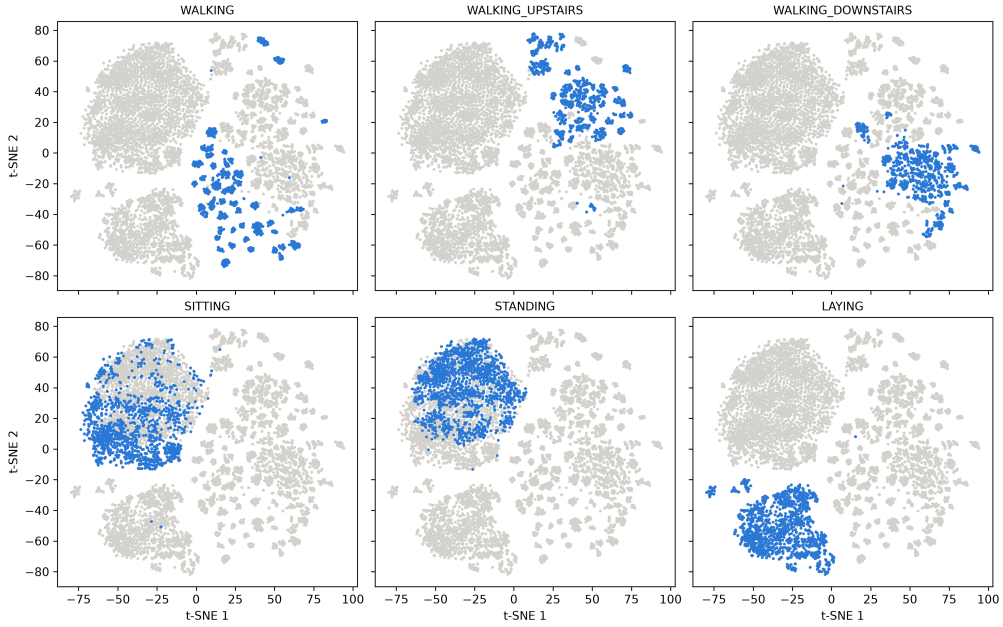


Figure 9: t-SNE embedding of the 540 distinct features; the axes have no meaning.

- `fBodyAcc-std()-X` separates the static from the dynamic activities: mean $\in [-0.99, -0.96]$ vs. $[-0.32, 0.12]$.
- `tGravityAcc-mean()-X` separates `LAYING`: mean $= -0.375$ vs. $[0.875, 0.941]$ for the other activities.
- Sitting and standing overlap.

- t-SNE, Figure 9:
  - Laying, the walking activities and the other static activities form distinct regions.
  - Sitting and standing share one region, partly separated.
  - Walking splits into many small, tight clusters.

References

[AGO+13] Davide Anguita, Alessandro Ghio, Luca Oneto, Xavier Parra, and Jorge Luis Reyes-Ortiz. A public domain dataset for human activity recognition using smartphones. In *21st European Symposium on Artificial Neural Networks, ESANN 2013, Bruges, Belgium, April 24-26, 2013*, 2013.